Multiplying polynomials

1

a

i y^2 **ii** $3y$ **iii** $7y$ **iv** 21

b $y^2+10y+21$

2

a $ab-6a+9b-54$

b $28rs+8r+49s+14$

c $x^2+7x+10$

d $x^2+12x+35$

e $v^2+18v+80$

f $x^2+3x-10$

g x^2-5x+6

h $m^2-3m-18$

i $5m^2-7m-24$

j $35w^2+39w+10$

k $42y^2+6y-36$

l $4c^2-56.2c+2.8$

m $\frac{1}{9}y^2+\frac{1}{36}y-\frac{1}{24}$

n $-90y^2+26y+48$

o $9y^2+90y+144$

p $7x^2-42x-189$

q $-x^2-7x-10$

r $-5y^2-45y-100$

s $-4y^2+12y+112$

t $-75x^2+110x-40$

u $6x^2-12x-18$

v $6x^{10}+13x^5y^4+5y^8$

w $-30x^2-42x+34$

x $-8x^2+40x+72$

y $18x^2-\frac{69x}{2}-8$

z $-4x^2-6x+32$

3 $\left(12x^2+144x+384\right)\ \mathrm{m}^3$

4 $42x^2+59xy+20y^2$

5

a $(h+9)\ \mathrm{cm}$ **b** $(w+8)\ \mathrm{cm}$ **c** $\left(4h+\frac{9w}{2}+36\right)\ \mathrm{cm}^2$

6 $\left(128y^2+48y\right)\ \mathrm{cm}^3$

7

a $5a^3 + 8a^2 - 2a + 4$

b $u^3 + u^2 - 15u - 20$

c $-25a^3 - 50a^2 - 45a - 20$

d $20c^4 - 15c^3 + 2c^2 + 6c - 4$

e $6a^3 - 13a^2b - 2ab^2 + 12b^3$

f $-12m^3 - 20m^2 + 8$

g $-56x^2 + 24xy - 12x + 36y + 108$

8 True

9 -13

10

a 12 b -18 c Yes

11

a 4 b 5

12 $12t^7$

13

a Yes b No c No d No

14

a No b Yes c Yes d No

e Yes f No

15

a Yes b No c Yes d No

16

a No b Yes c Yes d Yes

17 $\left(8y^3 + 22y^2 + 17y + 3\right)\ \text{ft}^3$

18

a $(10 - 2x)\ \text{cm}$ b $(6 - 2x)\ \text{cm}$ c $\left(4x^3 - 32x^2 + 60x\right)\ \text{cm}^3$

Additional questions

19

a 4 and -28

b 1 and -11

c Answers may vary. 2, 6, and 18 is one possibility. 3, 4, and 18 is another.

20

a The new dimensions are $(5+2x)$ and $(7+2x)$

b $4x^2+24x+35$

c $4x+24$

21

a The new dimensions are $(12-2x)$, $(12-2x)$, and $(5.5-2x)$

b Jerome will have a volume of $-4x^3+59x^2-276x+396$ left to fill.

22

a $(10x)(10x)=100x^2$, but Anette put $100x$ instead. Her final answer should have been $100x^2+20x+30x+6=100x^2+50x+6$

b

	x^2	$10x$	2
$10x$	$10x^3$	$100x^2$	$20x$
3	$3x^2$	$30x$	6

c $10x^3+103x^2+50x+6$

23 The result will be a polynomial with degree 7.

24

a Agree. The maximum number of terms that can be produced by multiplying two binomials is $2 \times 2 = 4$ if no terms combine.

b Answers will vary. A monomial multipled by a five term polynomial will always results in a five term polynomial. Another possible soluton is a binomial multipled by a trinomial that results in a $4th$ degree polynomial with exactly one pair of terms combining.
For example, $(x^2+3)(x^2-2x+1)=x^4-2x^3+2x^2-6x+3$